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AXITEC

AXIbiperfect GXXL TS
570 - 600 Wp

High performance bifacial solar module
144 halfcell, glass/glass, N-Type TOPCon

ALMM Enlisted Products	Equivalent Products
KNACK-NG2G-16B-144-570	AC-570TGB/144TS
KNACK-NG2G-16B-144-575	AC-575TGB/144TS
KNACK-NG2G-16B-144-580	AC-580TGB/144TS
KNACK-NG2G-16B-144-585	AC-585TGB/144TS
KNACK-NG2G-16B-144-590	AC-590TGB/144TS
KNACK-NG2G-16B-144-595	AC-595TGB/144TS
KNACK-NG2G-16B-144-600	AC-600TGB/144TS

ALMM Declaration

- The AXITEC Products mentioned above are manufactured by Knack Energy Pvt. Ltd.: at following address covered under the ALMM list: Survey Number 291 & 292, Besides Kamla Amut Industrial Estate, Near Hitachi Johnson, Budasan, Kadi, Mahesana, Gujarat - 384440, India.
- Knack Energy Products mentioned above are both BIS approved and ALMM enlisted.

This product has been manufactured under the technical supervision of AXITEC. For further information on warranty and installation procedures, please visit www.axitecsolar.com

Warranty by
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Survey Number 291 and 292, Besides Kamla Amut Industrial Estate, Near Hitachi Johnson, Budasan, Kadi, Mahesana, Gujarat - 384440 | www.knackenergy.com

Knack
Knack Energy Pvt. Ltd.
Empowering the future...

Authorized OEM Partner

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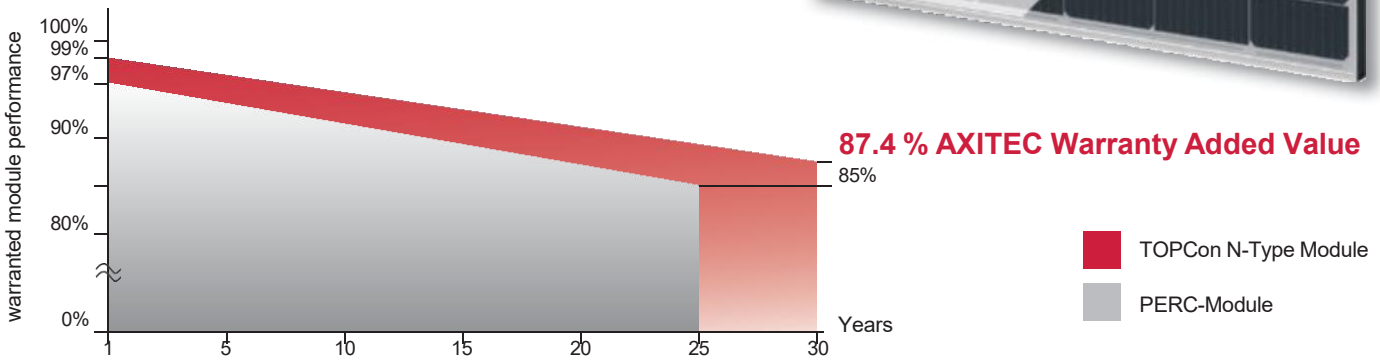
Fig. similar 144TGBEN231030A

The advantages:

- 30 years Manufacturer‘s warranty
- 30 years Performance guarantee
- Up to 30 % more power output by Bifacial-Technology
- More performance through innovative N-Type TOPCon-Technology
- PID reduced through glass/glass-Technology
- Increased safety through improved fire protection
- Guaranteed positive power tolerance from 0-5 Wp by individual measurement

Exclusive linear AXITEC high performance guarantee!



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Electrical data (at standard conditions (STC) irradiance 1000 watt/m². spectrum AM 1.5 at a cell temperature of 25°C)

Type	AC-570 TGB/144TS	AC-575 TGB/144TS	AC-580 TGB/144TS	AC-585 TGB/144TS	AC-590 TGB/144TS	AC-595 TGB/144TS	AC-600 TGB/144TS
Nominal output	**570 Wp**	**575Wp**	**580 Wp**	**585 Wp**	**590 Wp**	**595 Wp**	**600 Wp**
Nominal voltage Vmpp	44.74 V	44.95 V	45.15 V	45.35 V	45.55 V	45.75 V	45.95 V
Nominal current Impp	12.76 A	12.81 A	12.86 A	12.91 A	12.96 A	13.01 A	13.06 A
Short circuit current Isc	13.47 A	13.52 A	13.57 A	13.62 A	13.67 A	13.72 A	13.77 A
Open circuit voltage Voc	52.20 V	52.40 V	52.60 V	52.80 V	53.00 V	53.20 V	53.40 V
Module Conversion Efficiency	22.07%	22.26%	22.45%	22.65%	22.84%	23.03%	23.23%

Bifacial output - Backside Power gain

5% Power output	599 W	604 W	609 W	614 W	620 W	625 W	630 W
Module Efficiency	23.19%	23.38%	23.57%	23.77%	24%	24.20%	24.39%
15% Power output	656 W	661 W	667 W	673 W	679 W	6.84 W	690 W
Module Efficiency	25.40%	25.59%	25.82%	26.05%	26.28%	26.48%	26.71%
25% Power output	713 W	719 W	725 W	731 W	738 W	744 W	750 W
Module Efficiency	27.60%	27.83%	28.06%	28.30%	28.57%	28.80%	29.03%

Design

Front side	2.0 mm hardened, low-reflection white glass
Back side	2.0 mm hardened glass, cell spaces transparent
Cells	144 N-Type TOPCon bifacial high efficiency cells
Frame	35 mm silver aluminium frame

Mechanical data

L x W x H	2278 x 1134 x 35 mm
Weight	32.5 Kg kg with frame

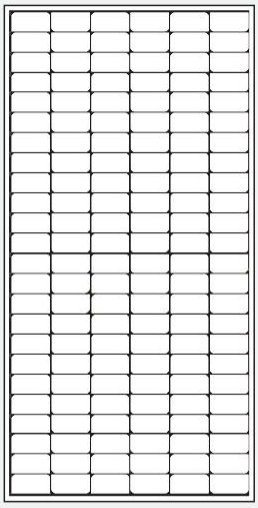
Mechanical load

Design load (pressure/suction)	3600 Pa / 1600 Pa *
Test load (pressure/suction)	5400 Pa / 2400 Pa *

* depending on the type of installation according to the installation instructions

Power connection

Socket	Protection Class IP68, 3 bypass diodes
Wire	approx. 300mm, 4 mm²
Plug-in system	MC4 compatible



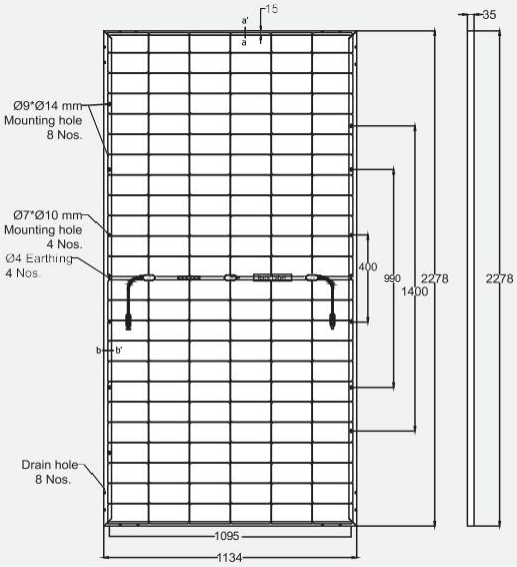


Fig. principle sketch

All dimensions in mm

Limit values

System voltage	1500 VDC
NOCT (nominal operating cell temperature)*	45°C +/-2K
Reverse current feed IR	30.0 A
Permissible operating temperature	-40°C to 85°C / -40F to 185F
Bifaciality	80 % ± 5 %
Fire class / Protection class	C (UL790) / II

(No external voltages greater than Voc may be applied to the module)

* NOCT. irradiance 800 W/m²; AM 1.5; wind speed 1 m/s; Temperature 20°C

Temperature coefficients

Voltage Voc	-0.26 %/K
Current Isc	0.046 %/K
Output Pmpp	-0.31 %/K

Low-light performance (Example for AC-580TGB/144TS)

I-U characteristic curve	Current Ipp	Voltage Vpp
200 W/m²	2.74 A	41.58 V
400 W/m²	5.54 A	42.06 V
600 W/m²	8.27 A	42.38 V
800 W/m²	10.94 A	42.75 V
1000 W/m²	13.43 A	43.20 V

Packaging

Module pieces per pallet	31
Pallet/Container (40' HC)	20
Module pieces per HC-container	620

Technical data are subject to change without prior notice. errors excepted. The measurement tolerances are +/-3%. Please be aware: All technical data provided in our data sheets are property of Axitec Energy GmbH & Co. KG and intended for information purposes for our customers only. We cannot accept any guarantee of completeness or accuracy. These data are prohibited for any kind of commercial use.